Sahaj Bhadja

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Summary

Software engineer with 2+ years building distributed backend systems, cloud data workflows, and SQL-backed services. Strong in reliability, scalable APIs, and production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant data services on Azure SQL and Blob Storage with normalized schemas and row-level controls, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous processing workflows with FastAPI background workers, supporting 1K+ AI jobs through structured job-state transitions, failure isolation, reliable recovery, and execution across concurrent production workloads.
- Designed ETL-style integration pipelines across FastAPI, SQL Server, PostgreSQL, and cloud storage, processing 5K+ monthly jobs with validation checks, reconciliation logic, and CI/CD-backed releases for dependable operations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with a microservice architecture, ensuring consistent data flow while supporting scalable service communication, distributed validation workflows, and maintainable backend components across multiple environments.
- Optimized large-scale API and ETL integrations using Python, SQL, AWS Glue, and S3, improving response time by 30% while enhancing pipeline efficiency by 68% across production data workflows.
- Maintained 20+ automated tests and CI pipelines with Jenkins, PyTest, and Mockito, strengthening release reliability by 68% and improving debugging, deployment consistency, and operational confidence across distributed builds.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed concurrent batch-processing workflows and secure file-ingestion APIs for an AI grading platform, handling 10,000+ daily requests while maintaining stable backend behavior, reliable execution, and scalable request processing.
- Implemented production backend services on AWS with CI/CD-driven releases, reducing feedback turnaround by 60% while supporting high-throughput workflows, multimodal processing, and dependable API performance under peak usage.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Automated Docker-based build and test workflows with Jenkins, cutting release time by 50% while improving deployment consistency, troubleshooting discipline, and shared engineering practices across distributed environments and teams.
- Built MERN platform features with REST APIs and WebSockets, supporting synchronized records across modular workflows while translating requirements into reliable end-to-end production behavior for shared internal applications.

Skills

Languages: Python, SQL, JavaScript, TypeScript, Java, C++, Bash
Backend: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, microservices, async workers
Data & Cloud: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, AWS, Azure, GCP, S3, AWS Glue, Blob Storage
Delivery & Systems: Docker, Jenkins, GitHub Actions, CI/CD, distributed systems, async processing, cloud architecture, system design, scalability, reliability, fault isolation, OOP, unit testing, debugging, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, GCP*
- Engineered a cloud-based ingestion and indexing platform processing 50K+ records, implementing incremental updates, metadata persistence, and performance tuning to reduce retrieval latency by 40% in production workflows.

Linguistix Code | *Python, AWS, Docker*
- Developed and deployed a containerized RESTful data processing API on AWS with automated CI/CD releases, validation checks, and production-grade reliability, demonstrating scalable service delivery across cloud environments.

FinNexus Live | *TypeScript, K8s, MongoDB*
- Built a scalable fintech platform supporting 50,000+ users and 1M+ transactions, combining real-time updates, secure APIs, and deployment automation to maintain responsive behavior under high-volume workloads.